12.3.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 12, Lesson 4: Solving Problems With Lengths of Two Tangents

Benchmark

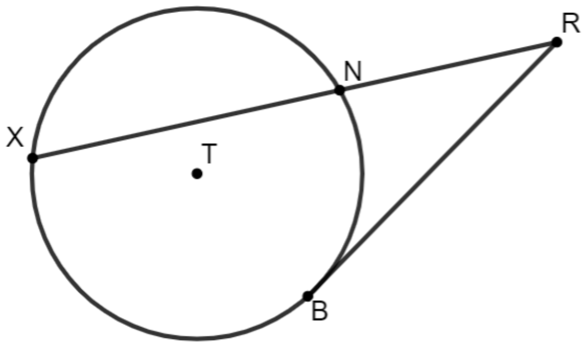
MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

- ☐ I can solve mathematical and real-world problems that involve lengths formed by the intersection of two tangents.

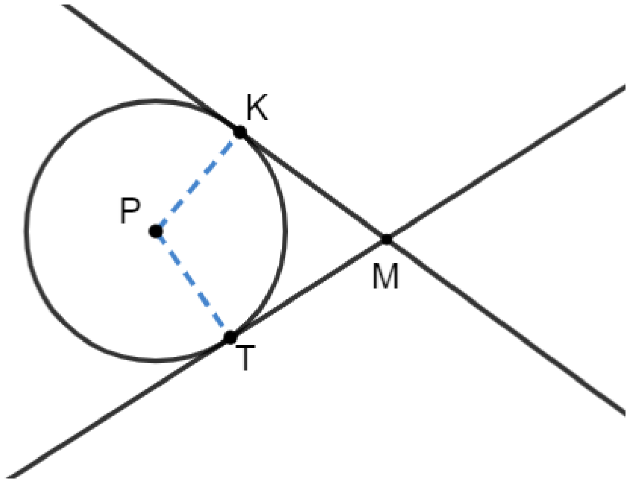
12.4.1 Warm-Up: Bell Work

1. Circle T has points X, N, and B on the circle. Secant $\overline{XR}$ and tangent $\overline{BR}$ intersect at point R outside the circle. Given that $XN = 16$ and $NR = 9$, find the length of $\overline{BR}$.



12.4.2 Exploration: Intersecting Tangent Lines

1. Circle P is shown with tangents $\overleftrightarrow{MK}$ and $\overleftrightarrow{MT}$ and radii PK and PT.



   a. Draw a dotted line from point P to point M.

   b. Discuss with your partner the relationships that can be determined given that $\overleftrightarrow{MK}$ and $\overleftrightarrow{MT}$ are tangents, and $\overline{PK}$ and $\overline{PT}$ are radii.

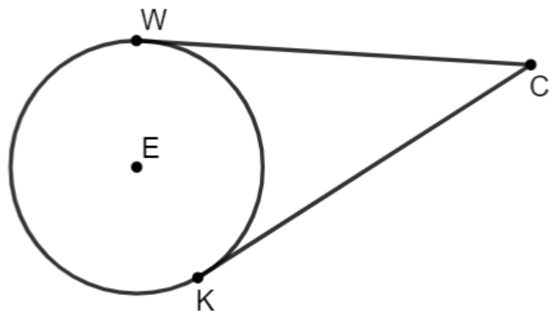
c. Discuss with your partner why HL congruence can be used to show that ΔMKP and ΔMTP are congruent.

d. Since $\Delta MKP \cong \Delta MTP$, what is true about $\overline{MK}$ and $\overline{MT}$?

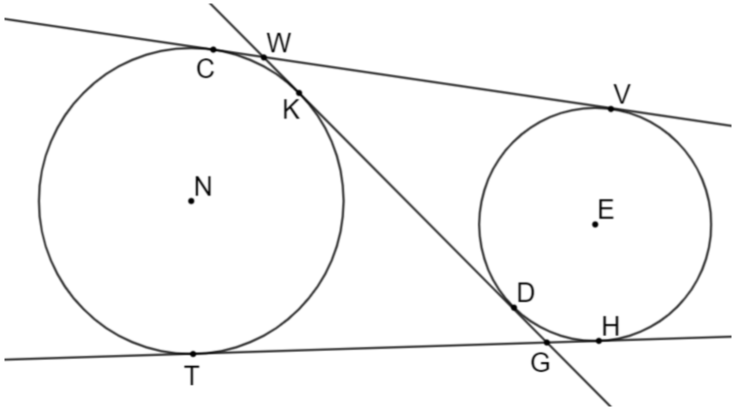
Tangent segments to a circle from the same point are congruent.

12.4.3 Guided Practice: Solving Problems Involving Intersecting Tangent Lines

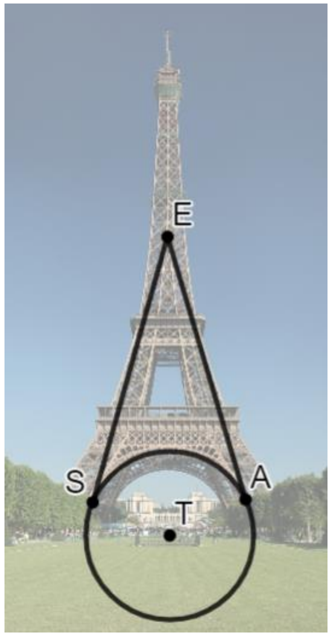
1. Circle E has points W and K on the circle. Tangents $\overline{WC}$ and $\overline{KC}$ intersect at point C outside the circle. If $WC = x + 15$ and $KC = 2x - 5$, find the length of $\overline{KC}$.



2. Circle N has points C, K, and T on the circle. Circle E has points V, H, and D on the circle. Tangents $\overline{TG}$ and $\overline{KG}$ intersect outside the circle at point G. Tangents $\overline{VW}$ and $\overline{WD}$ intersect outside the circle at point W.
If $TG = (7x - 36)$, $KG = (5x - 14)$, $VW = (8y + 13)$, and $WD = KG$, what is the value of y?

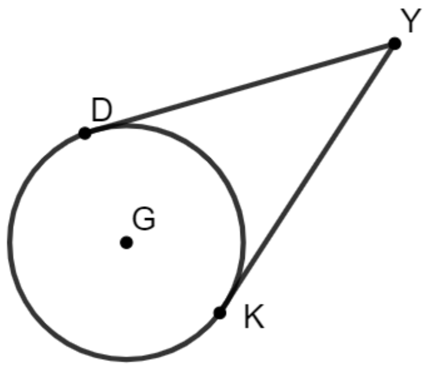


3. The Eiffel Tower in Paris, France stands at 324 meters tall. The tower has four semicircular arches at its base. Circle T represents one of the four arches at the base of the tower, where points S and A lie on the circle. If $SE = (17x + 27)$ meters (m) and $EA = (20x - 15)$ m, what is the length of $\overline{EA}$ in meters?

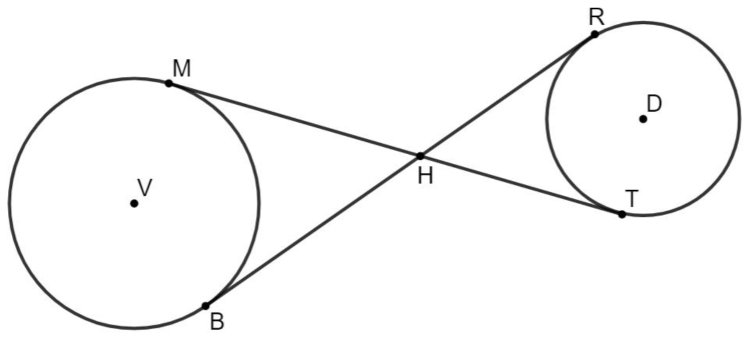


12.4.4 Your Turn!

1. Circle G has points D and K on the circle. Tangents $\overline{DY}$ and $\overline{KY}$ intersect at point Y outside the circle. If $DY = 4x - 19$ and $KY = 15$, find the value of x.



2. Points M and B are on the circle V and points R and T are on the circle D. $\overline{MT}$ and $\overline{BR}$ intersect at point H. $\overline{MT}$ is tangent to circle V at point M and tangent to circle D at point T. $\overline{BR}$ is tangent to circle V at point B and tangent to circle D at point R. If $MH = (4x - 8)$, $BH = (2x + 1)$, $TH = (2x - 1)$, and $RH = (5y - 7)$, what is the value of y?



12.4.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.

Unit 12, Lesson 5: Pythagorean Theorem and Tangent Lines

Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

- ☐ I can use the Pythagorean Theorem to solve mathematical and real-world problems involving the length of a tangent segment to a circle.